

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1– Secure Infrastructure Architecture

Course Code:	CSA5803	Course Title:	DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO4 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1
Weightage:	30% of CIA	Total Marks:	100
CO 4 Statement:	Design Secure Infrastructure Architecture using DiagrammeR or Drawio.		
Student Name:	Syed Saniya Iqbal	Reg. No.:	192465043
Section / Batch:	CSE-Cybersecurity	Date:	18/8/26

Instructions to Students

- Design a Secure Infrastructure Architecture for the selected application/system using DiagrammeR or Draw.io.
- Include essential components such as Firewall, WAF, Load Balancer, Web/Application Server, Database, IAM, IDS/IPS, and Monitoring/Logging wherever applicable.
- Clearly represent network zones, data flow, security controls, and trust boundaries in the architecture.
- Ensure the diagram is original, professionally formatted, clearly labelled, and submitted along with its editable source file.

Question Type	Focus Area	Questions	Marks
Secure Infrastructure Architecture	Conceptual Understanding	Q1	Q1 –100
GRAND TOTAL:		100 Marks	

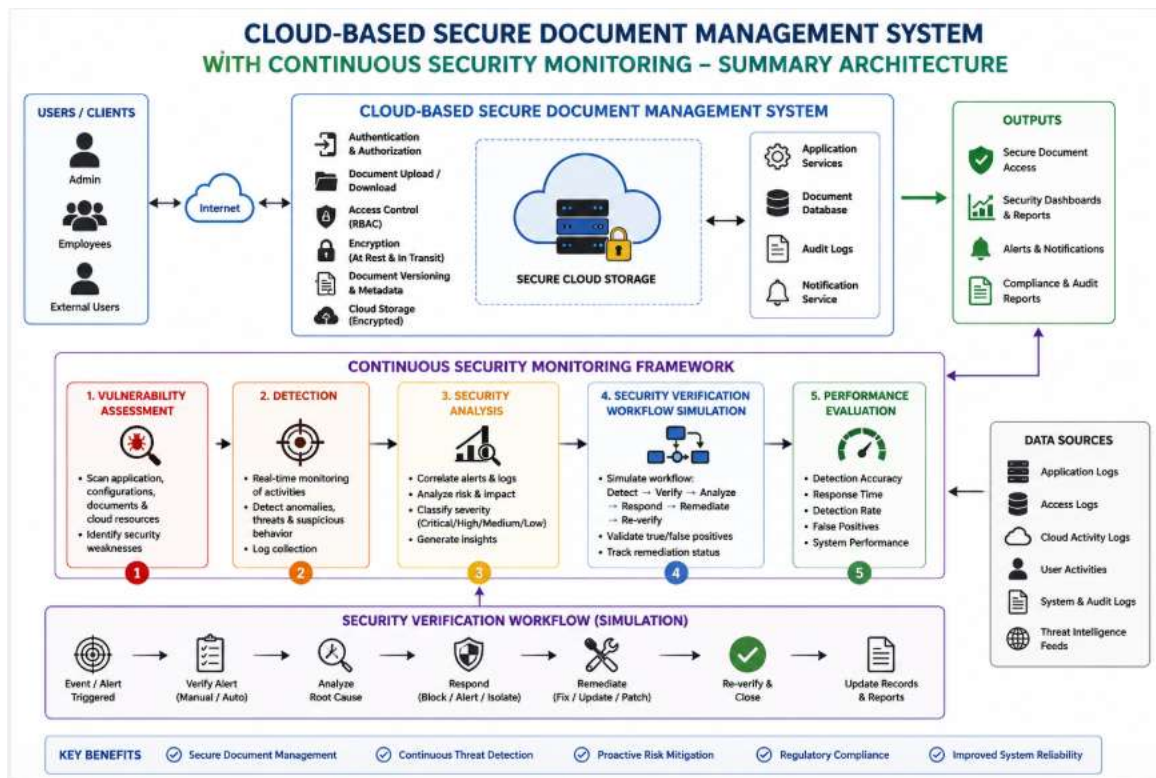
Code of Conduct

*I **Syed Saniya Iqbal(192465043)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **Syed Saniya Iqbal(192465043)**

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand



The architecture provides a **secure and scalable online examination environment**.

- **Students, invigilators, and administrators** access the system through the Internet.
- **Anti-DDoS, WAF, and firewalls** protect the system from external attacks.
- **Authentication and exam servers** handle login, exam delivery, and submission.
- **Proctoring services** continuously monitor student activity.
- **Network segmentation** separates public, application, database, and management networks.
- **Encrypted databases and backups** protect exam and student data.
- **Monitoring and SIEM** detect suspicious activity and generate alerts.
- **SOC** handles security analysis, incident response, and auditing.

Overall: The architecture ensures **secure access, attack protection, data confidentiality, continuous monitoring, and high availability** for the online examination system.